DP Biology – D3.3

Homeostasis

- How do humans keep internal conditions constant?
- Why is it useful for organisms to keep internal conditions constant?

Statements D3.3.1–6, SL and HL. Textbook pp. 760–767.
Image: goose bumps, one response to cold. EverJean, CC BY 2.0.



Objectives for each class

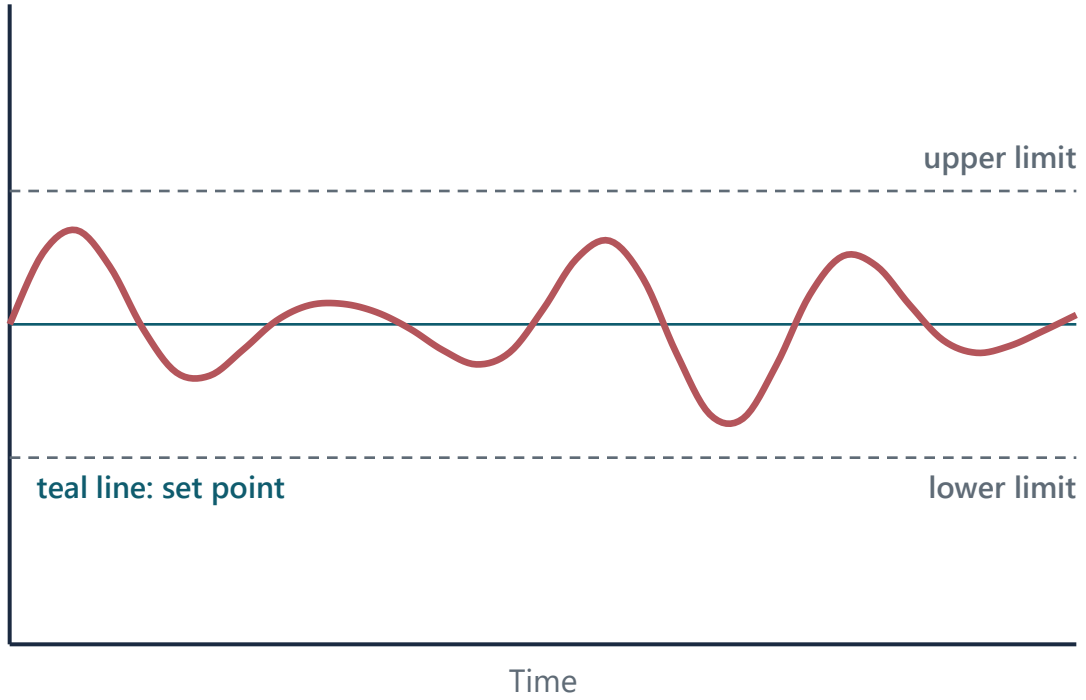
Class	Statements	By the end of the class, you can…
1	D3.3.1–3	Define homeostasis, explain negative feedback, and explain how insulin and glucagon control blood glucose
2	D3.3.4	Compare the two main types of diabetes, and interpret glucose tolerance data
3	D3.3.5–6	Explain how hormones and effectors regulate body temperature in humans

What homeostasis is

D3.3.1

Homeostasis keeps the internal environment steady: variables stay within **preset limits**, whatever happens outside.

Variable in humans	Typical value
Body temperature	about 37 °C
Blood pH	about 7.4
Blood glucose	about 90 mg per 100 cm^3
Blood water balance	narrow limits

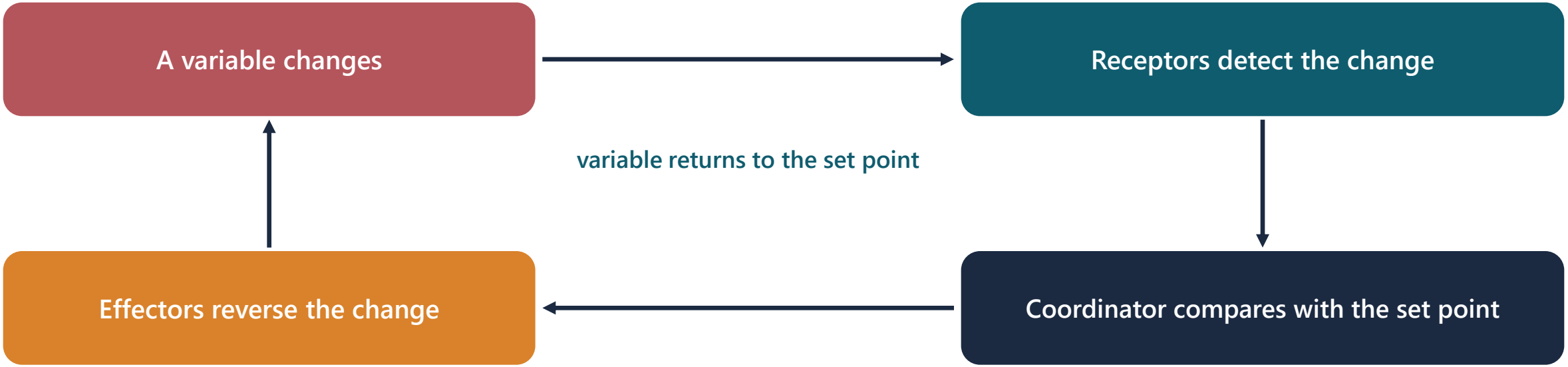


Common mistake

Homeostasis is about the internal environment, not a person changing their surroundings.

Negative feedback

D3.3.2



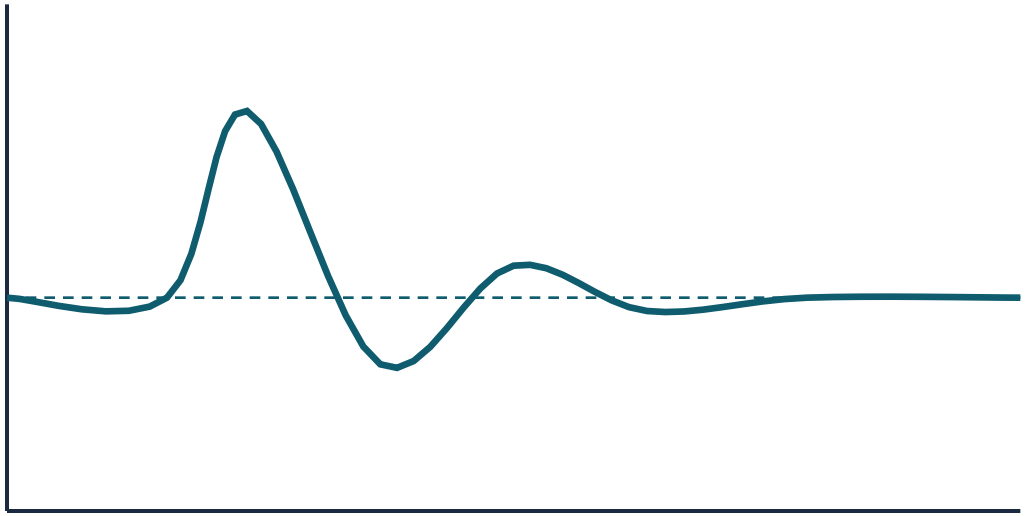
Negative feedback reverses any change, from above or below the set point. Once the variable is back, the effectors are switched off.

Example

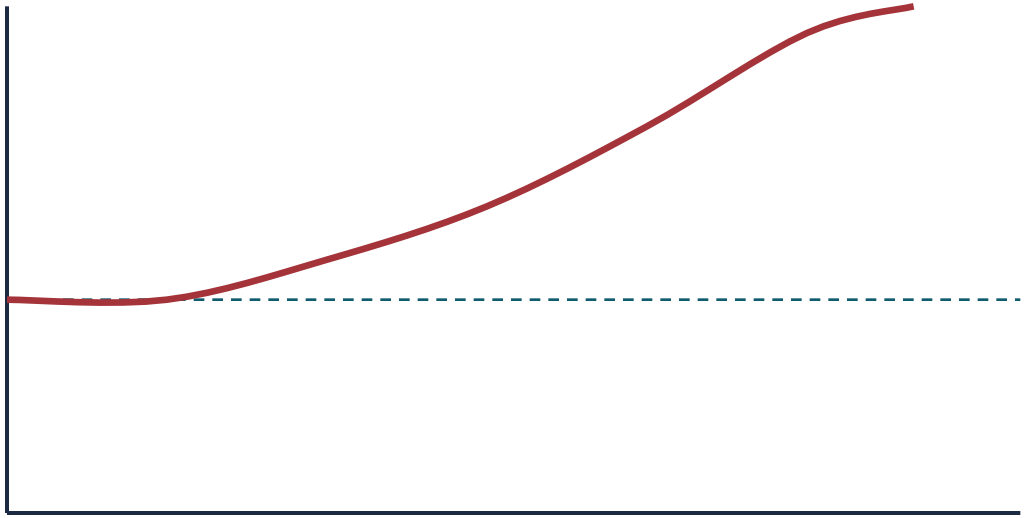
Too warm: skin thermoreceptors → hypothalamus → sweat glands → body cools.

Why negative, not positive, feedback?

D3.3.2



Negative feedback: back to the set point



Positive feedback: further and further away

Negative feedback **reverses** a change, so conditions stay stable. Positive feedback **amplifies** a change, so it cannot keep a variable steady.

Link

Positive feedback has uses: childbirth (D3.1) and fruit ripening (C3.1).

Why blood glucose must be controlled

D3.3.3

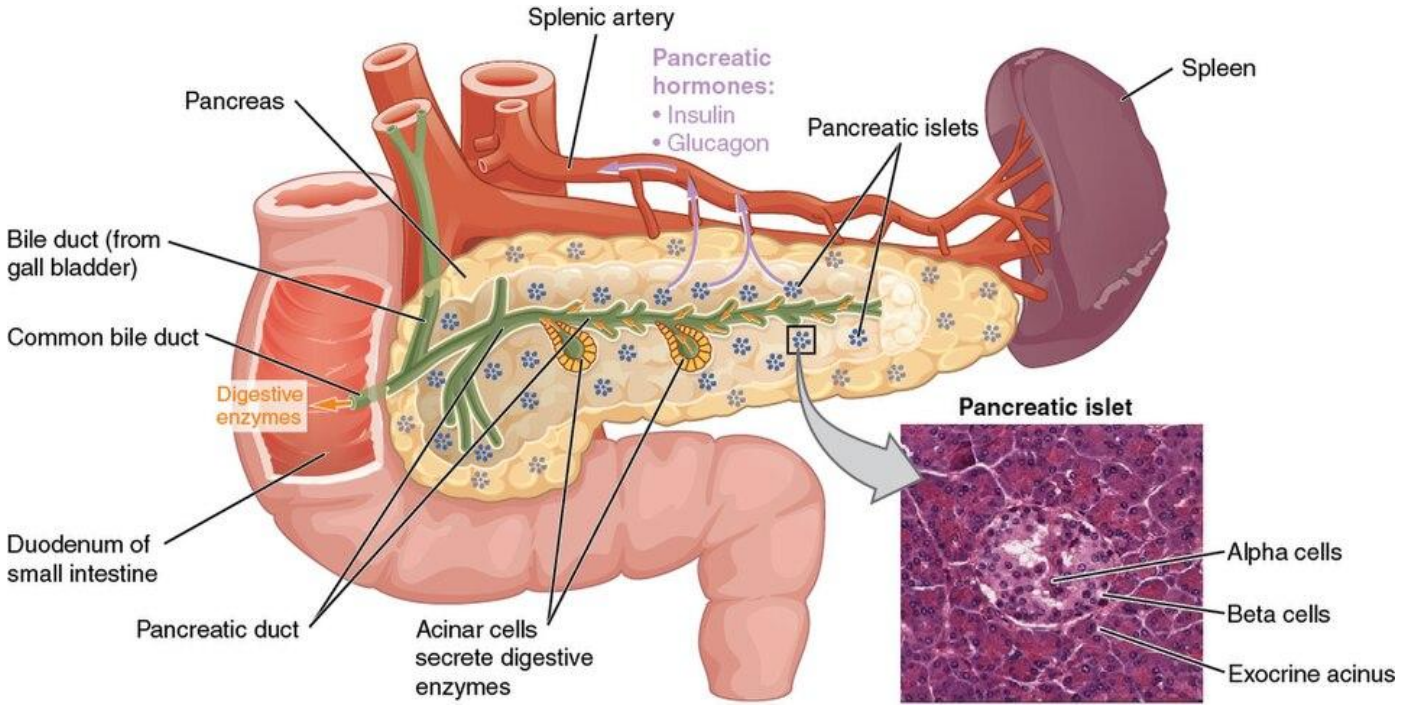
Condition	Blood glucose	Consequence
Too low	below about 70 mg	Brain short of fuel: fainting, coma
Normal	about 90 mg per 100 cm^3	Steady supply to cells
Too high	well above 150 mg	Water drawn from tissues; vessels damaged

Key idea

Brain cells store no glycogen, so they need a constant supply of glucose from the blood.

The pancreas makes two hormones

D3.3.3



The pancreas and its islets. OpenStax College, CC BY 3.0.

Hormone-secreting cells sit in the **islets of Langerhans**.

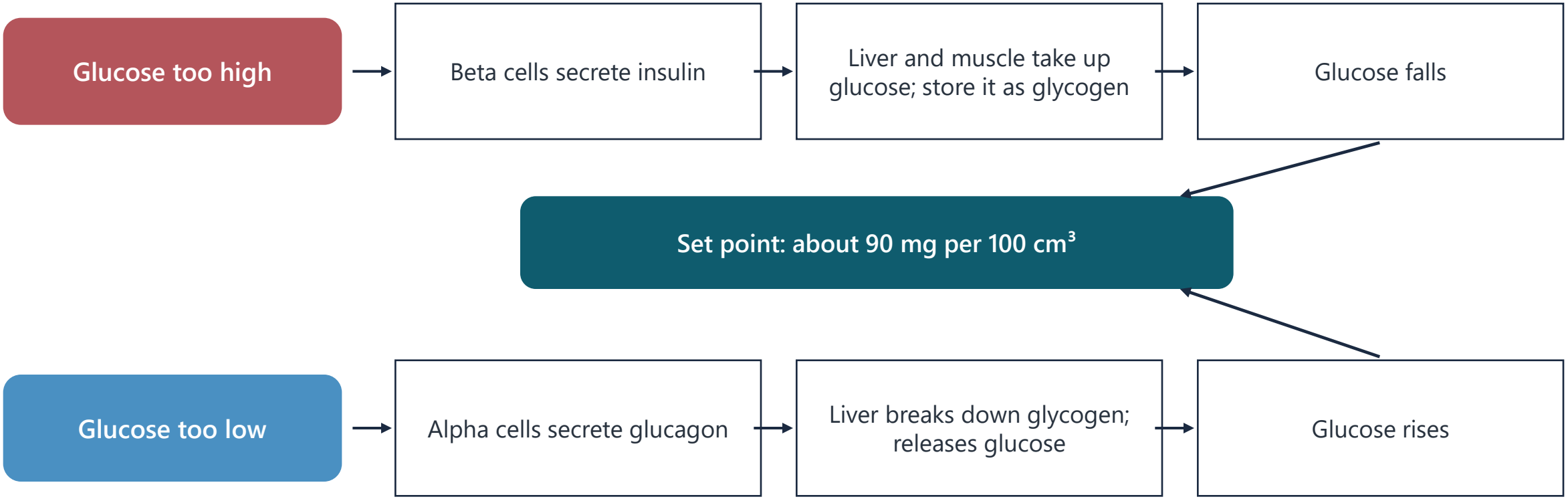
Beta cells secrete **insulin**. **Alpha cells** secrete **glucagon**.

Common mistake

The pancreas cells detect glucose themselves. Do not confuse glucagon (hormone) with glycogen (store).

Insulin and glucagon

D3.3.3



Common mistake

Glucagon is a hormone, not an enzyme: it signals liver cells, whose own enzymes break down glycogen.

Practice: predict the response

D3.3.3

After a large meal

Name the hormone, the cells that make it, and its effect on the liver.

A long run, no food

Explain how blood glucose is kept up.

A night's sleep

Predict glucagon and insulin levels at 7 a.m. and explain.

How to use this

Answer in two or three sentences each, naming receptor, hormone, target and effect.

Check-in: class 1

D3.3.1–3

1. Define homeostasis.
2. Explain why homeostasis uses negative feedback.
3. Name the cells that secrete insulin and glucagon.
4. Describe what happens when blood glucose rises after a meal.
5. Distinguish between glucagon and glycogen.

How to use this

Answer without your notes, then check against slides 3–9.

Two kinds of diabetes

D3.3.4

	Type 1	Type 2
What goes wrong	Beta cells destroyed: no insulin	Insulin made, but cells respond poorly
Cause	Immune system attacks beta cells	Target cells less sensitive to insulin
Usual onset	Childhood	Later life, now often younger
Share of cases	About 10%	About 90%

Key idea

Either way, blood glucose stays high after eating, because cells take up too little glucose.

Risk factors and prevention

D3.3.4

Type 1

- Genes
- Possibly a virus infection

Type 2

- Overweight
- Little exercise
- Age
- Family history

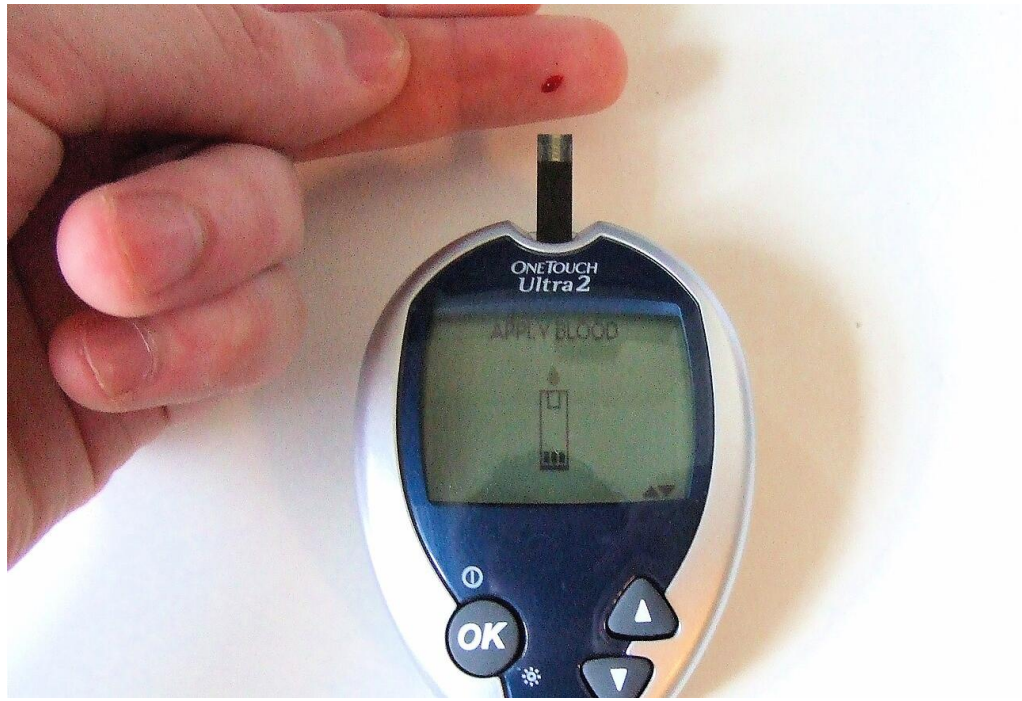
Type 2 is often **preventable**: a healthy diet, regular exercise and a healthy body mass all lower the risk. Type 1 cannot be prevented at present.

Think

Suggest why type 2 diabetes is now found in teenagers when it was once rare before 40.

Treating diabetes

D3.3.4



Monitoring blood glucose. David-i98, CC BY-SA 3.0.

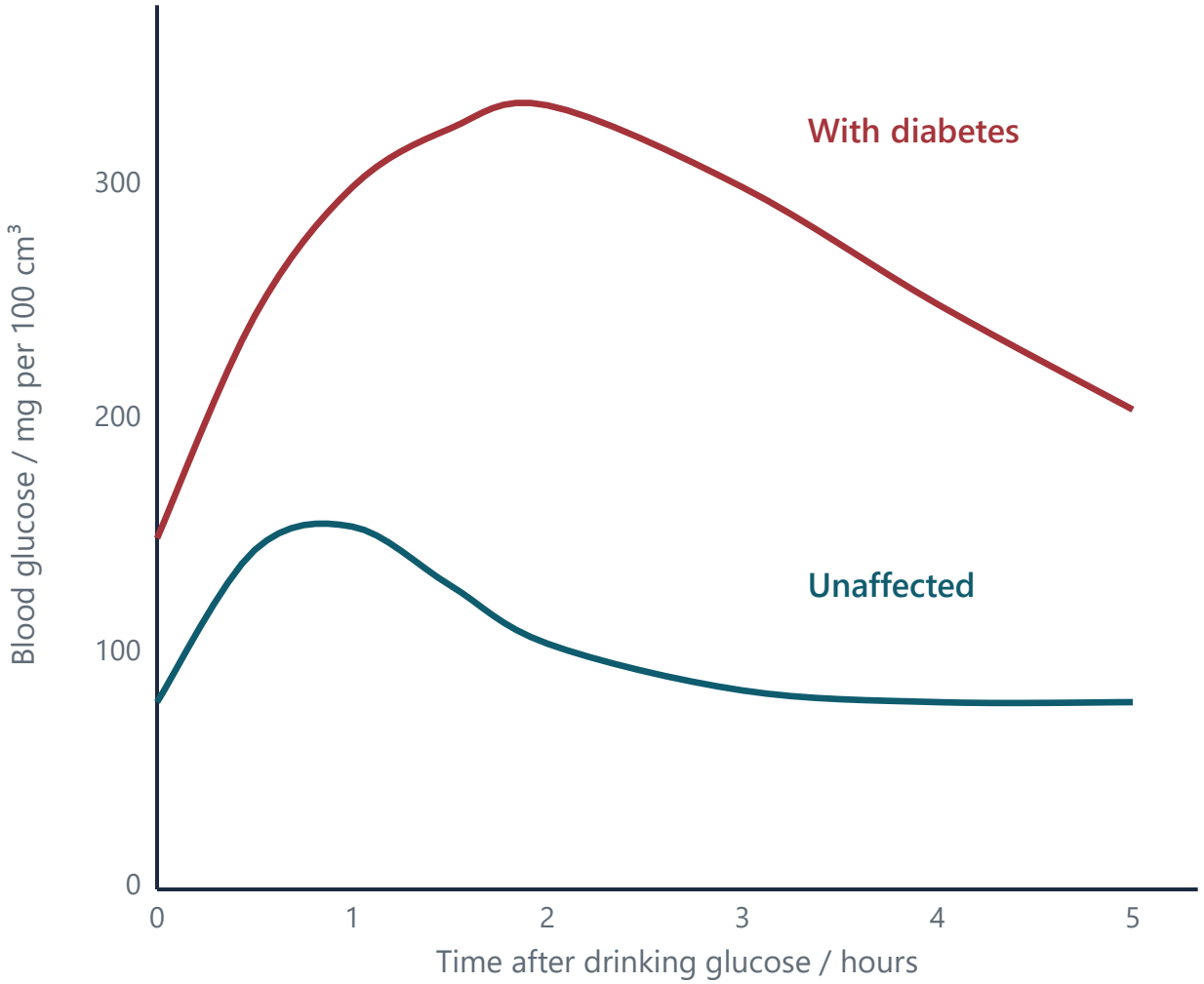
Type 1: regular **insulin** injections or a pump, matched to meals, with frequent glucose monitoring.

Type 2: diet and exercise first; medicines that improve the response to insulin; sometimes insulin later.

Why it matters

Long-term high glucose damages blood vessels, eyes, kidneys and nerves.

The glucose tolerance test



Invented values, based on a realistic pattern.

After fasting, the person drinks a set dose of glucose; blood glucose is measured for hours.

Unaffected: small rise, back to normal in 2–3 hours.
Diabetic: high at the start, a higher peak and a very slow fall.

Think

Why are samples taken at 0 and at 2 hours?

Working with data: glucose tolerance

D3.3.4

Blood glucose after drinking 75 g of glucose at 0 minutes. These data are invented for practice; the pattern is realistic.

Time / min	Person A / mmol dm^{-3}	Person B / mmol dm^{-3}
0	4.8	7.4
30	7.9	11.6
60	6.5	13.2
90	5.3	12.8
120	4.9	11.9

1. Identify which person may have diabetes, with evidence.
2. Explain the change in person A's glucose from 30 to 120 minutes.
3. Suggest why person B's glucose stays high.
4. Above 11.1 mmol dm^{-3} at 120 minutes suggests diabetes. Classify person B.

Check-in: class 2

D3.3.4

1. State the cause of type 1 diabetes.
2. Explain why blood glucose stays high in type 2 diabetes.
3. Give three risk factors for type 2 diabetes.
4. Compare how the two types are treated.
5. Describe the glucose tolerance curve of a person with diabetes.

How to use this

Answer without your notes, then check against slides 11–15.

Temperature control by negative feedback

D3.3.5

Thermoreceptors in the skin detect a change → Hypothalamus compares with the set point → Signals go to **effectors** → Heat loss or production changes → Temperature **returns** to about 37 °C



Heat loss in infrared: warm areas bright. NASA/IPAC, public domain.

The hypothalamus also has thermoreceptors that monitor **blood** temperature.

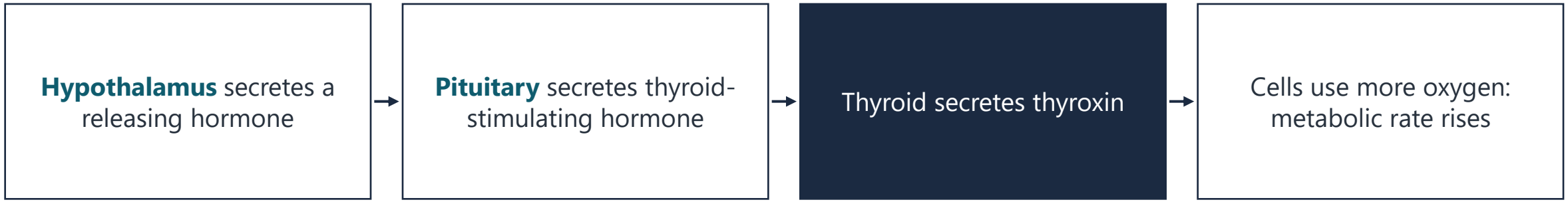
Most heat comes from metabolism; blood carries it around the body.

Effectors

Skeletal muscle, adipose tissue, skin arterioles, sweat glands and hair muscles.

Thyroxin controls heat production

D3.3.5



Effector tissue	Role
Skeletal muscle	Shivering releases heat
White adipose tissue	Insulates: a long-term response to cold
Brown adipose tissue	Releases heat fast by uncoupled respiration

High thyroxin reduces secretion by the hypothalamus and pituitary: negative feedback again.

Responses to heat and to cold

D3.3.6

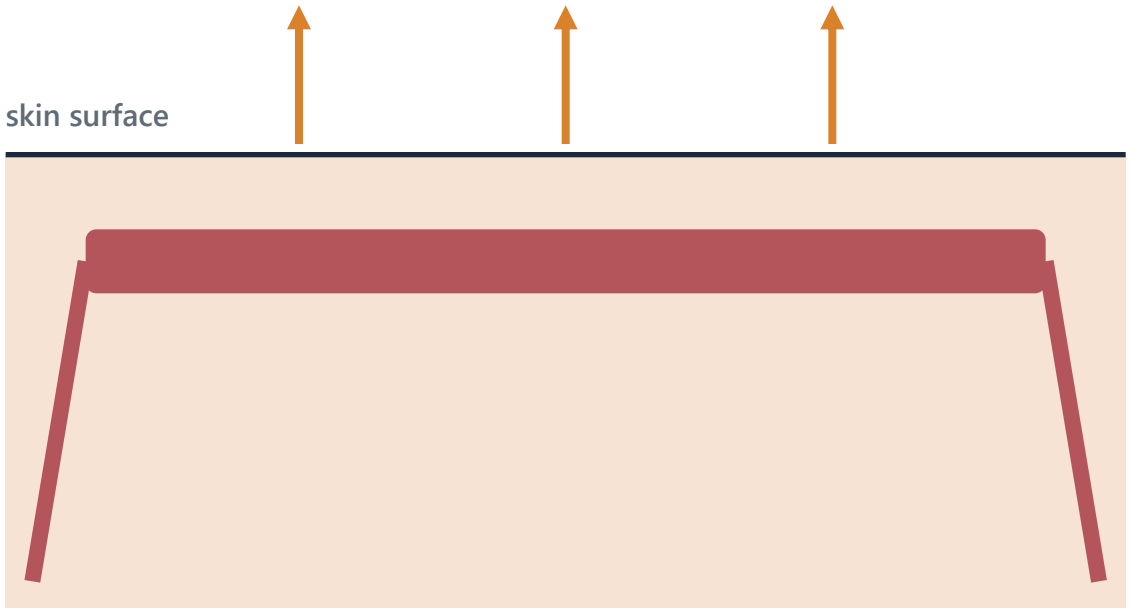
	Too hot	Too cold
Skin arterioles	**Vasodilation**: more blood to skin	**Vasoconstriction**: less blood to skin
Sweat glands	More sweat, which evaporates	Less sweat
Hair	Muscles relax: hairs flat	Muscles contract: hairs trap air
Muscle	No shivering	**Shivering** releases heat
Brown fat	Little activity	Uncoupled respiration releases heat

Common mistake

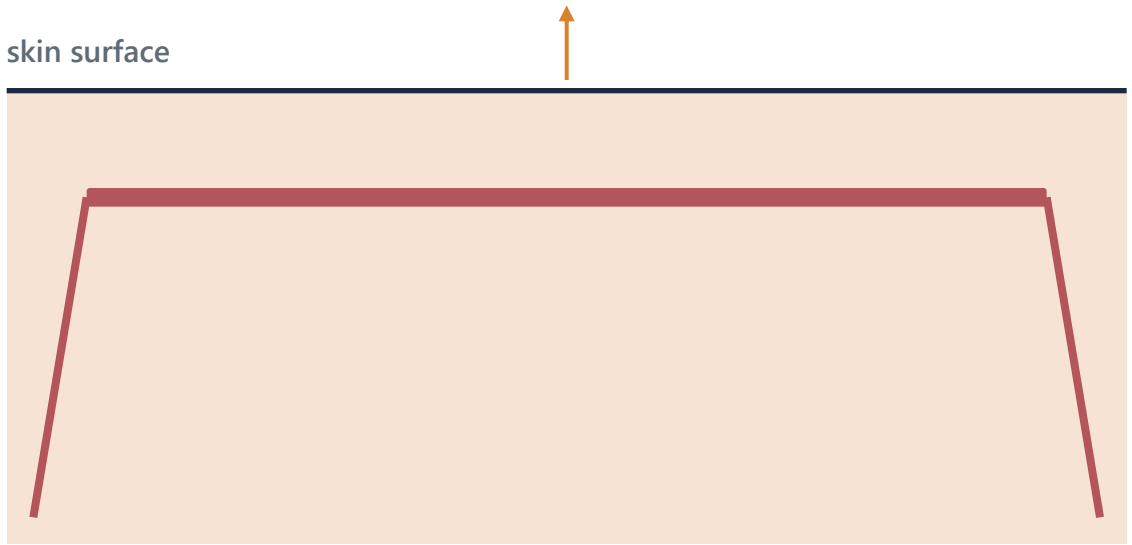
Blood vessels do not move up or down in the skin. Arterioles widen or narrow.

Vasodilation and vasoconstriction

D3.3.6



Hot: arterioles widen, more blood near the surface, more heat lost



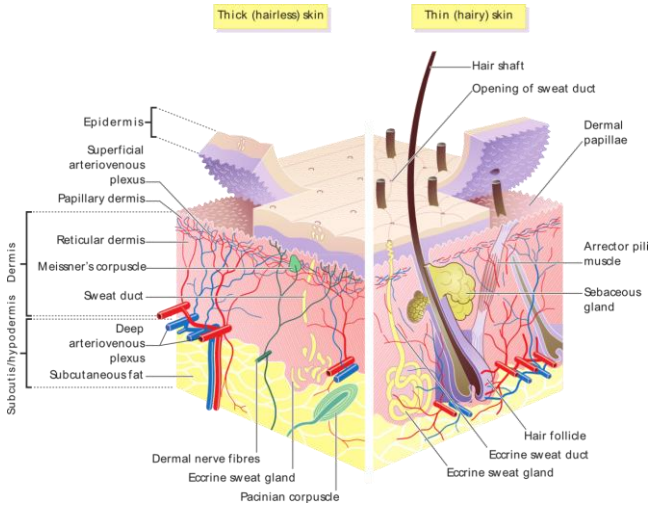
Cold: arterioles narrow, less blood near the surface, heat kept

Sweat cools the skin when it **evaporates**: breaking hydrogen bonds between water molecules takes heat from the skin.

Drawn diagram, not to scale. Orange arrows: heat lost from the skin.

Shivering and brown fat

D3.3.6



Skin, with sweat glands and hair muscles. Madhero88, CC BY-SA 3.0.

Shivering: fast, small muscle contractions. Respiration for them releases heat.

Uncoupled respiration in brown adipose tissue: mitochondria release energy as heat instead of making ATP. Newborn babies rely on it, as they lose heat fast.

Beyond humans

Birds and mammals also use behaviour, such as seeking shade. Detail is needed only for humans.

Working with data: a cold room

D3.3.5–6

A volunteer sat in light clothing in a room at 5 °C. The values are invented for practice; the pattern is realistic.

Time / min	Core / °C	Skin / °C
0	37.0	33
10	36.9	27
20	36.9	24
30	36.8	22

1. Compare the change in core and skin temperature.
2. Explain why skin temperature fell so much.
3. Explain how core temperature was kept almost steady.
4. Predict what happens if the volunteer stays much longer.

Why keep conditions constant?

D3.3.1

Enzymes work close to their optimum, so metabolism stays efficient. Cells get a steady supply of glucose and keep their water balance.

Mammals can stay active over a wide range of outside temperatures, so they can live in more places.

Discussion

A mammal must eat far more than a reptile of the same size to stay warm. Weigh up the costs and benefits of a constant body temperature.

Check-in: class 3

D3.3.5–6

1. Explain why thermoregulation is an example of negative feedback.
2. Outline the role of thyroxin.
3. Describe three responses to a rise in core temperature.
4. Explain how sweating cools the body.
5. Explain how brown adipose tissue releases heat.

How to use this

Answer without your notes, then check against slides 17–23.

Common mistakes to avoid

Not this	But this
Insulin turns glucose into glycogen.	Insulin signals liver and muscle cells; their enzymes make glycogen.
Glucagon and glycogen are the same.	Glucagon is a hormone; glycogen is a store.
Blood vessels move closer to the skin when hot.	Arterioles dilate; they do not move.
Sweating cools by losing water.	Sweat cools when it evaporates.
Type 2 means no insulin is made.	In type 2, cells respond poorly to insulin.

Summary: homeostasis

D3.3.1–6

	Key idea
D3.3.1	Internal variables kept within preset limits
D3.3.2	Negative feedback: receptor, coordinator, effector reverse a change
D3.3.3	Insulin (beta cells) lowers glucose; glucagon (alpha cells) raises it
D3.3.4	Type 1: no insulin. Type 2: poor response. GTT shows the difference
D3.3.5	Hypothalamus is the coordinator; thyroxin raises metabolic rate
D3.3.6	Vasodilation, sweating; vasoconstriction, shivering, brown fat

Check that you can...

1. **Define** homeostasis and **explain** negative feedback
2. **Explain** the control of blood glucose by insulin and glucagon
3. **Compare** the two main types of diabetes
4. **Interpret** glucose tolerance data
5. **Explain** thermoregulation as negative feedback
6. **Outline** the responses to heat and cold in humans

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain why homeostasis uses negative feedback and not positive feedback.

2. A person has just eaten a large meal. Name the hormone secreted, the cells that secrete it, and one effect on the liver.

3. Give two responses that the body makes when core temperature falls.

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